

SCHOOL OF ARTIFICIAL INTELLIGENCE

AWS Machine Learning Engineer Nanodegree

Syllabus

Overview

Refine your machine learning skills with this AWS Machine Learning Engineer Nanodegree. Learn to deploy models on SageMaker and design automated workflows with AWS Lambda and Step Functions.

Nanodegree Program



Intermediate



94 hours



4.7 (57 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

AWS familiarity

Intermediate Python

API proficiency

Jupyter notebooks

Deep learning

Basic Python

Basic probability

You will also need to be able to communicate fluently and professionally in **written and spoken English**.

Skills You'll Learn

Introduction to Machine Learning

Feature engineering | Machine learning fluency | AI business context | Machine learning use cases | Data loading with sagemaker | Amazon s3 | Model training | Linear models | Xgboost | Autogluon | Pandas | Sagemaker studio notebooks | Model performance metrics | Tree-based models | Sagemaker ground truth | Machine learning lifecycle | Dataset annotation | Sagemaker jumpstart | Machine learning dataset fundamentals | scikit-learn | Automated machine learning | Sagemaker data wrangler | Hyperparameter tuning

Developing your First ML Workflow

Sagemaker processing | Sagemaker batch transform jobs | Sagemaker training jobs | Sagemaker clarify | Machine learning pipeline creation | Sagemaker pipelines | Model monitoring | Sagemaker model endpoints | AWS Step Functions | Sagemaker model monitor | Sagemaker feature store | AWS lambda

Deep Learning Topics with Computer Vision and NLP

Neural network basics | Sagemaker jumpstart | Machine learning framework fundamentals | Sagemaker training jobs | Transformer neural networks | Sagemaker debugger | Image classification | Training neural networks | Deep learning model optimization | Transfer learning | PyTorch | Model deployment with sagemaker | Convolutional neural networks | Text classification | Model performance metrics | Hyperparameter tuning

Operationalizing Machine Learning on SageMaker



Cloud resource allocation | AWS lambda | Distributed model training with sagemaker | Amazon elastic compute cloud
| Sagemaker feature store | Cloud security in AWS | Cloud cost management | Sagemaker logs | Cloud performance
management | AWS storage services | Training data manifest files | Sagemaker autoscaling | Vpc

Courses

01. Welcome to AWS Machine Learning Engineer Nanodegree

45 minutes

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| L1 | An Introduction to Your Nanodegree Program | Welcome! We're so glad you're here. Join us in learning a bit more about what to expect and ways to succeed. |
| L2 | Getting Help | You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content. |

02. Introduction to Machine Learning

18 hours

Gain foundational machine learning expertise using AWS SageMaker, from data preparation and exploratory analysis to deploying powerful models like XGBoost and AutoGluon. Covering the entire ML workflow, including feature engineering, model tuning, and evaluation, you'll gain practical experience with real-world data projects. Designed for those familiar with Python and data science basics, the curriculum equips you to handle diverse ML tasks confidently and effectively, making it ideal for anyone looking to apply machine learning in various professional and industry settings.

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| L1 | Introduction to Machine Learning | Overview of key background around Machine Learning and preparing you to be successful in the rest of this course. |
| L2 | Exploratory Data Analysis | Use AWS SageMaker Studio to access S3 datasets and perform data analysis, feature engineering with Data Wrangler and Pandas. And finally label new data using SageMaker Ground Truth. |
| L3 | Machine Learning Concepts | In this lesson you'll learn about ML Lifecycles, how to differentiate between supervised vs. unsupervised ML, regression methods, and classification methods. |
| L4 | Model Deployment Workflow | In this lesson you'll load a dataset, clean/create features, train a regression/classification model with scikit learn, evaluate a model and tune a model's hyperparameter. |
| L5 | Algorithms and Tools | In this lesson you'll train, test, and optimize on liner, tree-based, XGBoost, and AutoGluon Tabular models. And you will also create a model using SageMaker Jumpstart |



L6

Project:

Predict Bike Sharing Demand with AutoGluon

Train a model using AutoGluon to predict bike sharing demand, and see how highly you can place in the competition!

03. Developing your First ML Workflow

15 hours

Step into the world of ML engineering with interactive AWS projects. Train models, deploy endpoints, and automate workflows using SageMaker, Lambda, and Step Functions to deliver efficient machine learning applications.

L1

Introduction to Developing ML Workflows

This lesson gives an introduction to the course, including prerequisites, final project, stakeholders, and tools & environment.

L2

SageMaker Essentials

This lesson will go over SageMaker essential services such as training jobs, endpoints, batch transforms, and processing jobs.

L3

Designing Your First Workflow

This lesson will discuss machine learning workflows and AWS tools such as Lambda, Step Function for building a workflow.

L4

Monitoring a ML Workflow

This lesson will go over monitoring a machine learning workflow and some useful services within AWS to help you monitoring the healthy of data and machine learning models.

L5

Project:

Project: Build a ML Workflow For Scones Unlimited On Amazon SageMaker

In the project, you will build and ship an image classification model with AWS SageMaker for Scones Unlimited, a scone-delivery-focused logistic company.

04. Deep Learning Topics with Computer Vision and NLP

15 hours

In this course you will learn how to train, finetune and deploy deep learning models using Amazon SageMaker.

You'll begin by learning what deep learning is, where it is used, and the tools used by deep learning engineers. Next we will learn about artificial neurons and neural networks and how to train them. After that we will learn about advanced neural network architectures like Convolutional Neural Networks and BERT as well as how to finetune them for specific tasks. Finally, you will learn about Amazon SageMaker and you will take everything you learned and do them in SageMaker Studio.

L1

Introduction to Deep Learning Topics within Computer Vision and NLP

In this lesson, we will give a background around Deep Learning for Computer Vision and NLP and preparing you to be successful in the rest of this course.



L2	Introduction to Deep Learning	In this lesson, you will learn about neural networks, cost functions, optimization, and how to train a neural network.
L3	Common Model Architecture Types and Fine-Tuning	In this lesson you will learn about Model Architectures, Convolutions, and Fine-tuning.
L4	Deploy Deep Learning Models on SageMaker	In this lesson, you will learn how to apply all you have learned about deep learning in AWS SageMaker.
L5	Project: Image Classification using AWS SageMaker	In this project, you will use AWS SageMaker to finetune a pretrained model and perform a image classification using profiling, debugging, and hyperparameter tuning.

05. Operationalizing Machine Learning on SageMaker

18 hours

This course covers advanced topics related to deploying professional machine learning projects on SageMaker. Students will learn how to maximize output while decreasing costs. They will also learn how to deploy projects that can handle high traffic, how to work with especially large datasets, and how to approach security in machine learning AWS applications.

L1	Introduction to Operationalizing Machine Learning on SageMaker	In this introductory lesson, we will give you a course overview of topics and design. We will also introduce what exactly operationalizing machine learning means as well as how it applies.
L2	Manage Compute Resources in AWS Accounts to Ensure Efficient Utilization	This lesson is about managing computing resources effectively. We'll talk about lowering costs and getting more with less.
L3	Train Models on Large-Scale Datasets Using Distributed Training	This lesson is about training models on large datasets. We'll talk about distributed models, distributed data, and some skills related to distributed training.
L4	Construct Pipelines for High Throughput, Low Latency Models	This lesson is about high throughput, low latency models. Essentially, this means that we'll be talking about preparing your projects to deal with high traffic and minimal time delays.
L5	Design Secure Machine Learning Projects in AWS	Our final lesson is about security. Security is crucial for all major machine learning projects, so these skills can be very helpful in your career.
L6	Project: Operationalizing an AWS ML Project	Your goal in this project will be to use several important tools and features of AWS to adjust, improve, configure, and prepare the model you started with for production-grade deployment.



06. Capstone Build Your Own Machine Learning Portfolio

28 hours

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| L1 | Machine Learning Engineer Capstone Project | Put your Machine Learning Engineer skills to the test by solving a real-world problem using all that you have learned throughout the program. |
| L2 | Project:
Capstone Proposal | Submit a Project Proposal for the Project that you selected. |
| L3 | Project:
Machine Learning Capstone | Once your project proposal is approved now it is time to actually build out the project. |

07. Congratulations!

10 minutes

Congratulations on finishing your program!

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| L1 | Congratulations! | Congratulations on your graduation from this program! Please join us in celebrating your accomplishments. |
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Meet Your Instructors



Matt Maybeno

Principal Software Engineer

Matt is a Principal Software Engineer at SOCi. With a masters in Bioinformatics from SDSU, he utilizes his cross domain expertise to build solutions in NLP and predictive analytics.



Bradford Tuckfield

Data Scientist and Writer

Bradford Tuckfield is a data scientist and writer. He has worked on applications of data science in a variety of industries. He's the author of Dive Into Algorithms, forthcoming with No Starch Press.



Soham Chatterjee

Graduate student at the Nanyang Technological University

Soham is an Intel® Software Innovator and a former Deep Learning Researcher at Saama Technologies. He is currently a Masters by Research student at NTU, Singapore. His research is on Edge Computing, IoT and Neuromorphic Hardware.



Charles Landau

Technical Lead, AI/ML - Guidehouse

Charles holds a MPA from George Washington University, where he focused on econometrics and regulatory policy, and holds a BA from Boston University. At Guidehouse, he supports data scientists and developers working on internal and client-facing ML platforms.



Joseph Nicolls

Lead Engineer, ML/AI

Joseph has a bachelor's degree in Biomedical Computation from Stanford University and over 5 years of technical experience in developing scalable machine learning workflows for cybersecurity use cases. In his free time, he enjoys reading with his wife and jogging by the San Francisco Bay.

Why Udacity



Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately



24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices